From K-Means Initialization to Topology-Aware Max-Cut

Complete research journey · methods, experiments, evidence and next tests

SDP-inspired vector factors · spherical clustering · local search · G-set

Experiments 001–023 · 29 October 2026

Prepared for Professor Ángel E. Rodríguez Fernández

Verified result at a glance — Experiment021

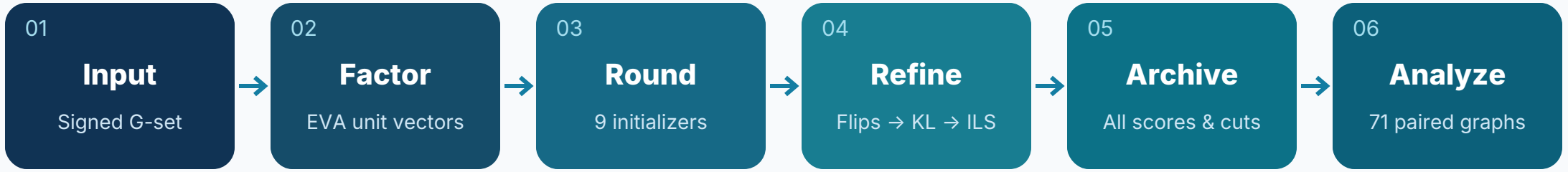
Scope	Verified result
Canonical G-set graphs	**71 / 71**
Initialization families	**9**
Matched restarts per graph and method	**50**
Fully archived Max-Cut searches	**31,950**
Missing factors / incomplete graphs	**0 / 0**
Graphs where best-of-nine beats best-of-control	**58 / 71**

Important: these are feasible **binary Max-Cut** results, **not** certified global optima or a general Max-k-Cut result for $k > 2$.

Source: Experiment021 report.

This counts Experiment021 only; do not add 1,207 scheduled Experiment023 cells to completed searches.

Research map: the verified experimental backbone



One fixed vector factor per graph · 50 matched restarts · immutable saved partitions

One graph, one immutable vector factor, multiple comparable rounding strategies, exact re-scoring and archived outputs.

1. What problem did we solve?

From a discrete graph partition to a geometric representation.

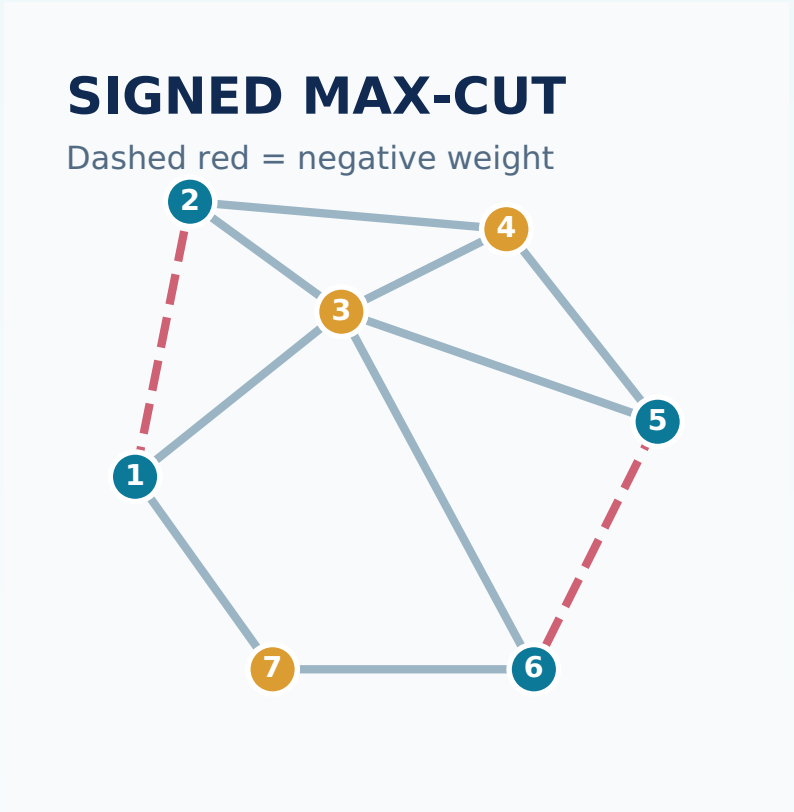
Formal objective: signed Max-Cut

For an undirected graph $G = (V, E, w)$, choose binary labels $c_i \in \{0, 1\}$:

$$\max_{c \in \{0,1\}^{|V|}} C(c), \qquad C(c) = \sum_{\{i,j\} \in E} w_{ij}\, \mathbf{1}[c_i \neq c_j].$$

- **Positive** w_{ij}: rewards placing endpoints on opposite sides.
- **Negative** w_{ij}: penalizes separating the endpoints.
- All reported cut scores use the **original signed weights**.

The clustering objective is **not** the cut objective: it proposes a partition that must subsequently be evaluated on the graph.



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Classical SDP relaxation (the theoretical reference)

Let $L = D - W$ and $X = VV^\top$ be a Gram matrix. For general $k \geq 2$:

$$\max_{X \succeq 0,\, X_{ii}=1} \frac{k-1}{2k}\mathrm{Tr}(LX), \qquad X_{ij} \geq -\frac{1}{k-1} \quad (k > 2).$$

At $k = 2$, the pairwise lower bound is redundant. Factorization represents each vertex i by a unit vector v_i.

Crucial distinction: the large-scale campaign used **rank-restricted nonconvex EVA factors**; these are not proven solutions or certified bounds for the full convex SDP.

Source: repository `AGENTS.md`, `RESEARCH_HANDOFF.md` and Rodríguez Fernández (2021, Chapters 6–7).

Goemans–Williamson: random hyperplane rounding

Draw $r \sim N(0, I_p)$ and set

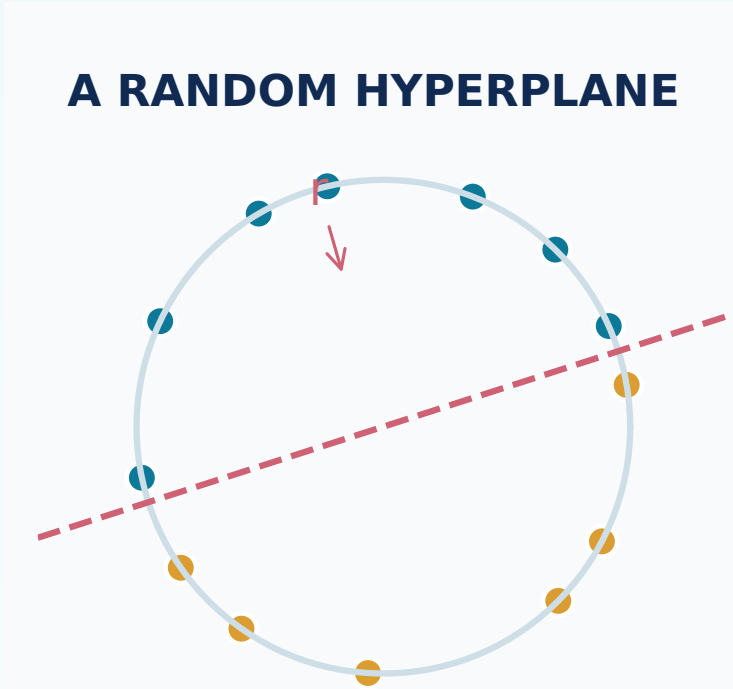
$$c_i = \mathbf{1}[v_i^\top r < 0].$$

The hyperplane $v^\top r = 0$ separates embedded vertices into two sides.

The starting question: why choose an arbitrary hyperplane if the vector embedding already contains structure from an optimization problem?

- Replace the initial two-way split with **spherical clustering**.
- Keep the original graph cut as the evaluation metric.
- Preserve the GW-equivalent starting cut in the original K-MeansNM control.

Source: Rodríguez-Fernández et al. (2020), Sections 1–3.



ILLUSTRATIVE GRAPH GEOMETRY

How spherical K-means enters Max-Cut rounding

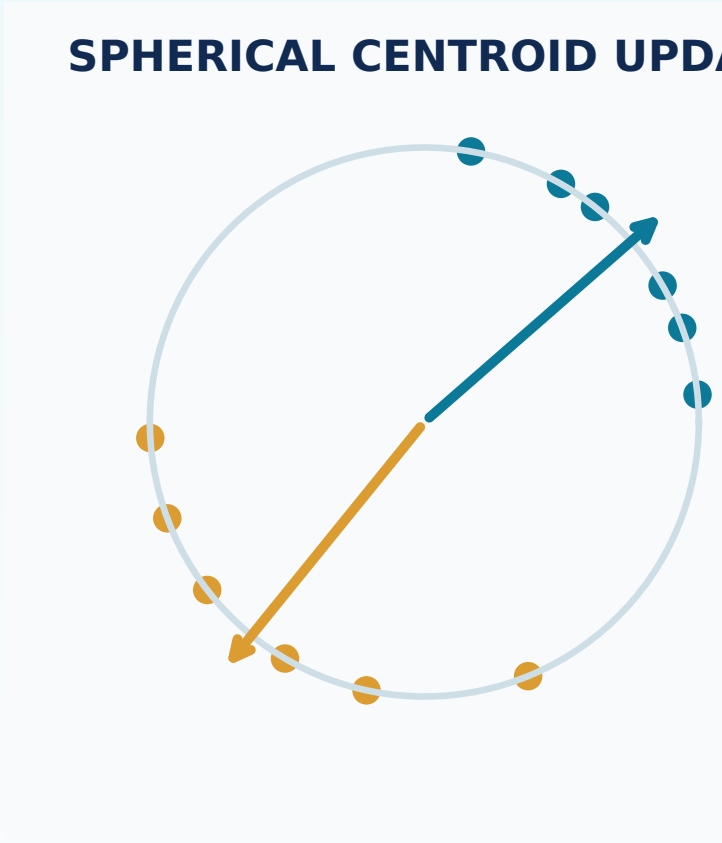
Original K-MeansNM initializes two unit centroids as u and $-u$. Its first assignment is the GW hyperplane partition.

Then alternate:

1. Assign each v_i to the centroid with maximum dot product.
2. Normalize each cluster's mean to update its centroid.
3. Keep the **best original-graph cut observed along the trajectory**.

Geometric distortion can decrease without the cut increasing at every step. This is why evaluating **intermediate** iterates matters.

Source: Rodríguez-Fernández et al. (2020), Eq. (4), Lemma 1 and Section 3.1.



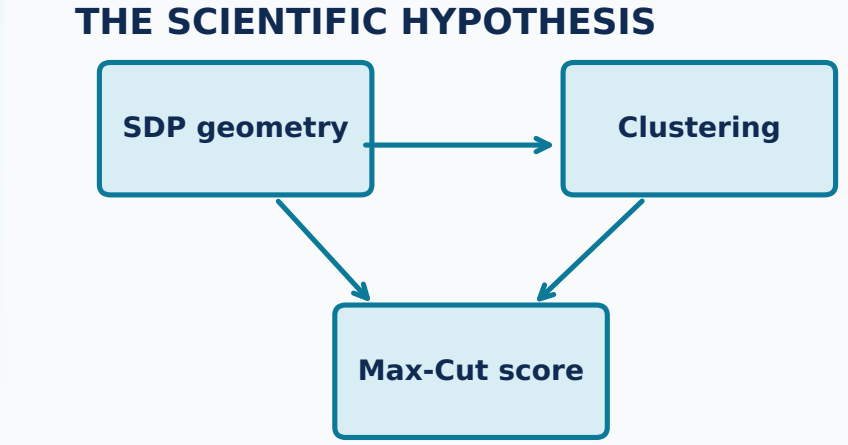
ILLUSTRATIVE GRAPH GEOMETRY

Three foundations from the literature

Work	What it contributed to our design
Rodríguez Fernández (2021), thesis	Max-Cut / Max-k-Cut, SDP geometry, fitness landscapes, clustering rounding
Rodríguez-Fernández et al. (2020)	K-MeansNM, K-Means2N and comparison with GW on G-set
Felzenszwalb, Klivans & Paul (2022)	Convex geometry of the k-way elliptope and iterative fixed-point rounding

The 2022 fixed-point method is **related prior art**, not a claimed component of our Experiment021 implementation.

Sources: project PDFs and research handoff.



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2. How the research evolved

Prototypes, failed hypotheses, stronger refiners and a controlled comparison.

Experimental chronology: from groundwork to a stable pipeline

Experiments	Research direction	What we learned
001–003	I/O, full SDP tests, EVA vector factors and checkpoints	Separate exact graph scoring from numerical factor optimization
005–009	MST partitions, balanced cuts, forest coarsening	Geometry-only bipartitions can be too restrictive
010–013	Generalized KL, Tabu, cut-aware rounding, pipeline grid, ILS	Post-rounding search is essential; historic G63 best **26,917**
015–016	Direct hyperplane optimization; soft Bernoulli / entropy approaches	Alternative searches did not replace the strongest pipeline
017–018	Multi-group clustering, exact binary merging; nine initializers	Fix the downstream pipeline and test initialization systematically
019–021	Matched G63 experiment, 68-graph expansion, 71-graph completion	Produce a complete auditable statistical dataset

Numbering note: unlisted experiment IDs are not asserted to be completed.

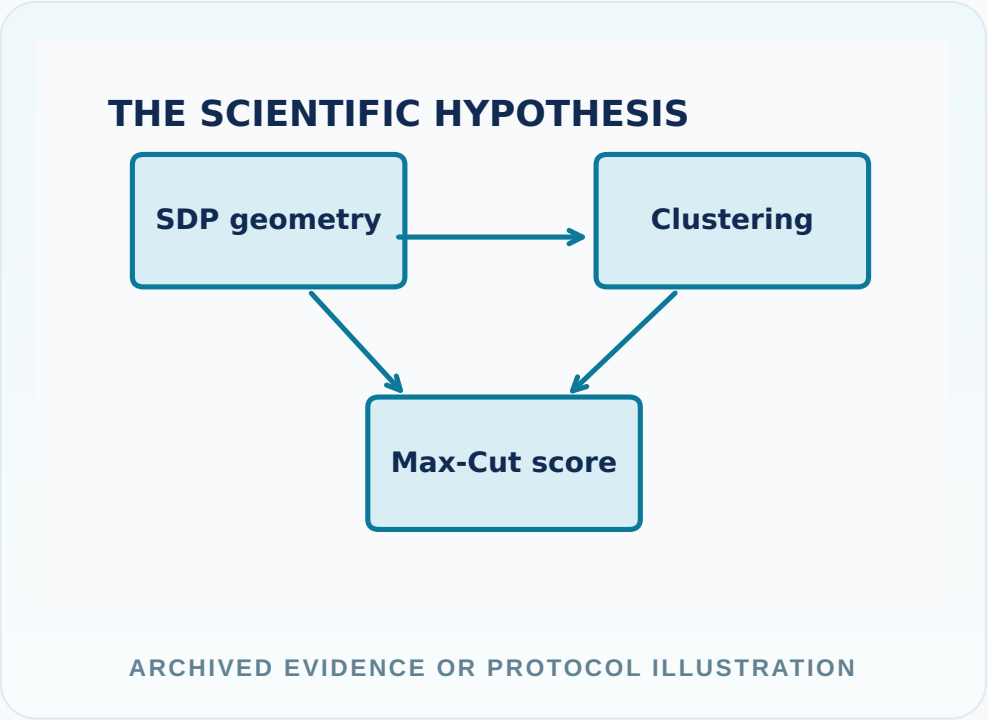
Source: `RESEARCH_HANDOFF.md` and the individual experiment notes.

Step 1 — Build a verifiable experimental substrate

- Keep canonical G-set text graphs, signed weighted edge lists and hashes.
- Implement numerical scoring separately from factor generation.
- Compare the small full-convex SDP pathway with scalable EVA approaches.
- Save reusable factors, iteration histories, seeds and binary partitions.
- Protect large jobs with explicit hardware/resource guards.

Why: we needed to distinguish a better *rounded cut* from a different embedding, input graph or solver approximation.

Repo: `src/maxkcut/{objective.py,sdp.py,eva.py,archive.py}`; Experiments001–003.



Step 2 — Obtain geometry with EVA

EVA optimizes a **rank-restricted vector relaxation** over row-unit embeddings $Y \in \mathbb{R}^{n \times p}$, with Y_i representing vertex i.

For the three final missing instances:

$$p = \lceil \sqrt{2(n+1)} \rceil = 119 \quad \text{for } n = 7000.$$

- Sequential Numba updates; one BLAS thread.
- Maximum 10,000 sweeps; angular tolerance **0.001 radians**.
- Seed **2026**; checkpoints every **25** sweeps.
- **No claim** of full-SDP convergence or global optimality.

Source: Experiment021 note and `.github/workflows/experiment021-complete-gset.yml`.

NOT INTERCHANGEABLE

Primal SDP	EVA factor
Convex full-rank relaxation	Nonconvex rank restriction

EVA convergence ≠ certified SDP bound

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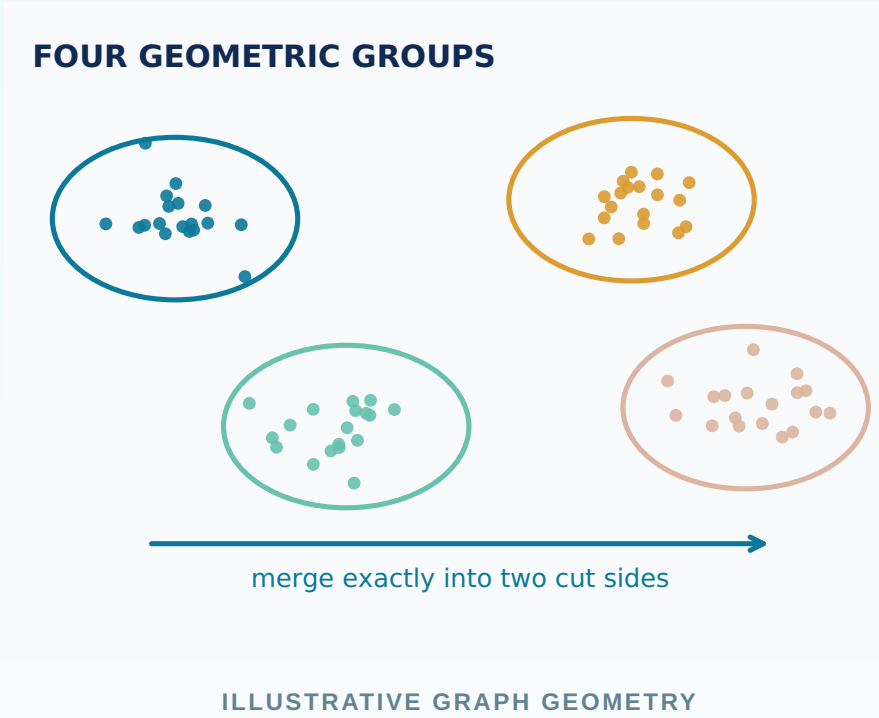
Step 3 — Test alternatives to naive rounding

MST approaches (005–009): full-dimensional Euclidean / other distances; tree-edge removals, multicomponent contractions, binary regrouping and vertex refinement.

Reason for moving on: rigid geometric clusters could miss good signed-graph cuts; G63's one-edge MST result was far below the later K-MeansNM + refinement baseline.

Critical invariant: when q MST edges are removed, the $q+1$ components are subsequently assigned back to **two** Max-Cut sides, not treated as $q+1$-Cut solutions.

Source: Experiment005 and Experiment009 notes.



Step 4 — Improve a cut with exact vertex gains

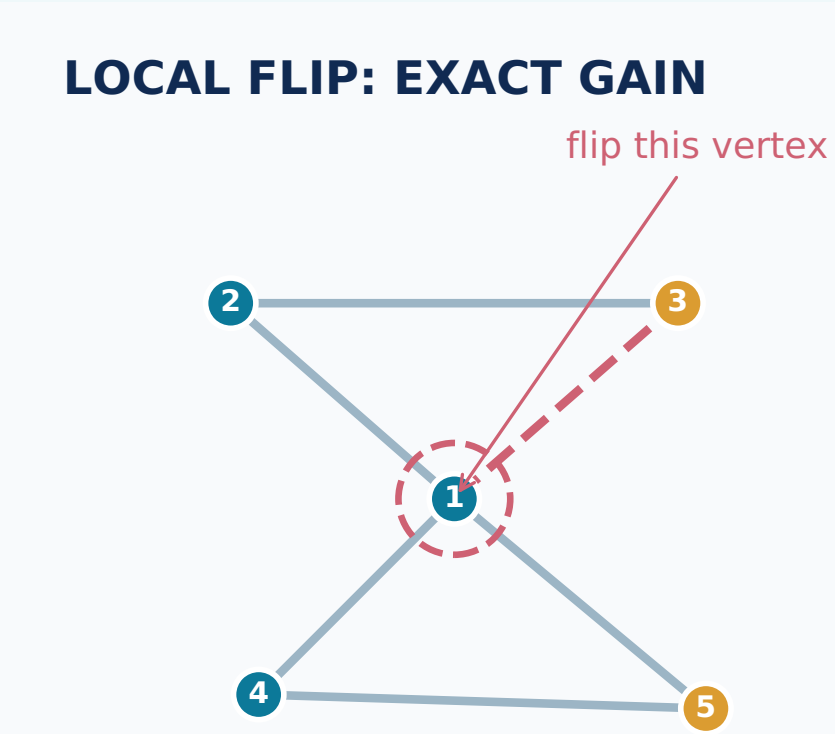
Represent side labels by spins $s_i \in \{-1, +1\}$:

$$C(s) = \frac{1}{2} \sum_{\{i,j\} \in E} w_{ij}(1 - s_i s_j), \qquad \Delta_i = s_i \sum_{j:\{i,j\} \in E} w_{ij} s_j.$$

Here Δ_i is the **exact change in cut** produced by flipping vertex i.

- **Flips:** accept positive-gain moves until locally stable or out of sweeps.
- Use the original signed sparse graph; update neighboring gains efficiently.
- Retain and independently rescore the resulting feasible partition.

Source: Experiment010 note, `src/maxkcut/` local-search implementation.



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Step 5 — Escape the one-flip local optimum

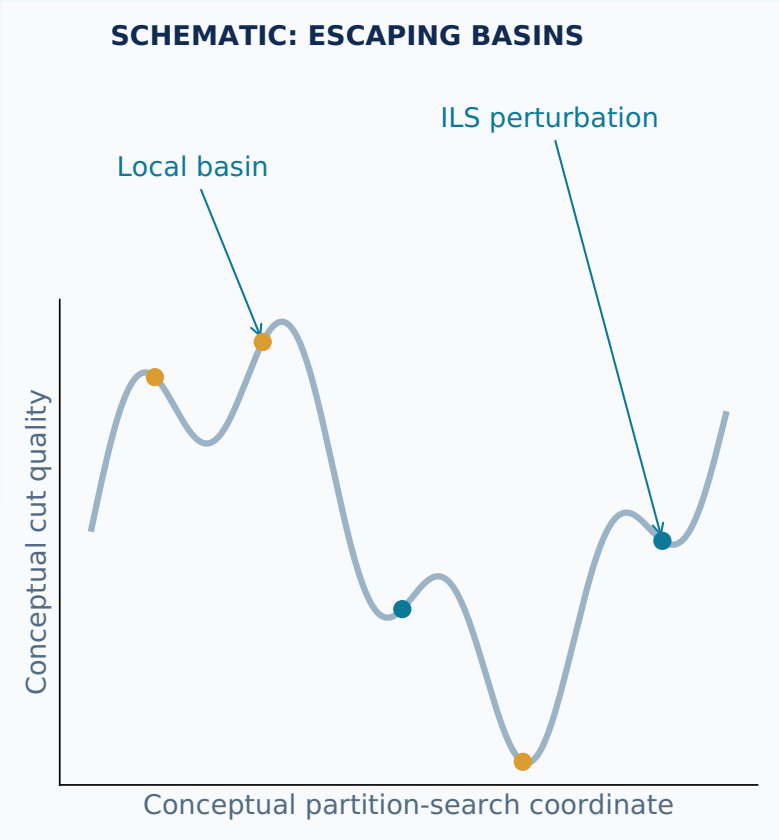
Generalized KL (not classical balanced pair-swapping):

1. In one pass, repeatedly choose the highest-gain unlocked vertex.
2. Permit temporarily negative gains.
3. Remember the best **strictly positive cumulative prefix**.
4. Accept that prefix; repeat up to the pass budget.

Iterated Local Search (ILS): perturb an elite partition, run inner KL, retain the best feasible cut visited.

This explains why the final score can be much stronger than the initial spherical clustering score.

Source: Experiments010 and 013; see Experiment019 pipeline.



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Step 6 — Select the full refinement pipeline (Experiment013)

```
Frozen EVA vector factor
        ↓
Round with candidate method
        ↓
Positive-gain vertex flips
        ↓
Generalized signed KL
        ↓
KL-based iterated local search
        ↓
Final feasible signed Max-Cut + archived partition
```

The historically best verified **G63 feasible cut was 26,917**. The exploratory **27,047** target remained 130 higher and was **not** a proven optimum.

Source: `configs/experiment013-grid.json`, Experiment013 note.

Step 7 — Explore other models, but preserve the main baseline

Experiment	Hypothesis tested	Methodological distinction
015	A better rounding normal may be found directly	Exact great-semicircle event sweeps; stochastic smoothed-gradient ascent
016	Soft partition distributions may escape hard-rounding limitations	Entropy-regularized Bernoulli assignments, CEM/Gibbs proposals
017	More than two clusters may reveal useful latent structure	2/4/6/8/10 groups, **exact** merge back to binary cuts
018	Initialization of two centers may be the main lever	Nine systematic initializer families

For Experiment017, q geometric groups can be merged to two labels by enumerating $2^{q-1}-1$ nontrivial complements; at $q=10$, **511** mergers.

These experiments motivate **controlled initialization comparisons**, not a claim that every exploratory strategy improved the G63 record.

3. The frozen comparison protocol

Nine initializers, one embedding per graph, identical downstream budgets.

The nine initialization methods: part I

Name	How the initial centroids were selected
Original paired K-MeansNM	Random unit vector u and antipode $-u$
Independent centers	Two independently drawn random unit directions
Sample antipodal	An actual embedding row Y_i and $-Y_i$
Sample pair	Two different randomly chosen embedding rows
Farthest pair	One sampled row and a row maximally distant by cosine

All methods operate on the same graph-specific, **fixed EVA factor**.

Source: Experiment018 note and Experiment021 methodology.

The nine initialization methods: part II

Name	How the initial centroids were selected
Spherical K-means++	First sampled row; second selected by distance-squared weighting
Mean perturbation	Two perturbations around the normalized embedding mean
PCA antipodal	Leading centered principal direction + jitter, then antipode
GW screen-16	Evaluate 16 random candidate hyperplanes and choose the best

Budget warning: GW screen-16 uses **16 additional graph-cut evaluations** per restart, and PCA has preprocessing costs. Equal restarts does **not** mean equal compute.

Source: Experiment018 note.

Identical downstream refinement for every method

Stage	Frozen setting
Spherical Lloyd	$k = 2$, up to **100** iterations; keep best actual cut
Positive vertex flips	Up to **20** sweeps
Generalized KL	Up to **30** passes
KL-based ILS	**80** cycles; **8** inner KL passes
ILS perturbation sizes	**8, 16, 32, 64** vertices
ILS temperature	**12**
Restarts	**50** per graph and method

All scores are recorded at **Round → Flip → KL → Final**.

Source: Experiments019–021.

Reproducible RNG design: matched downstream seeds

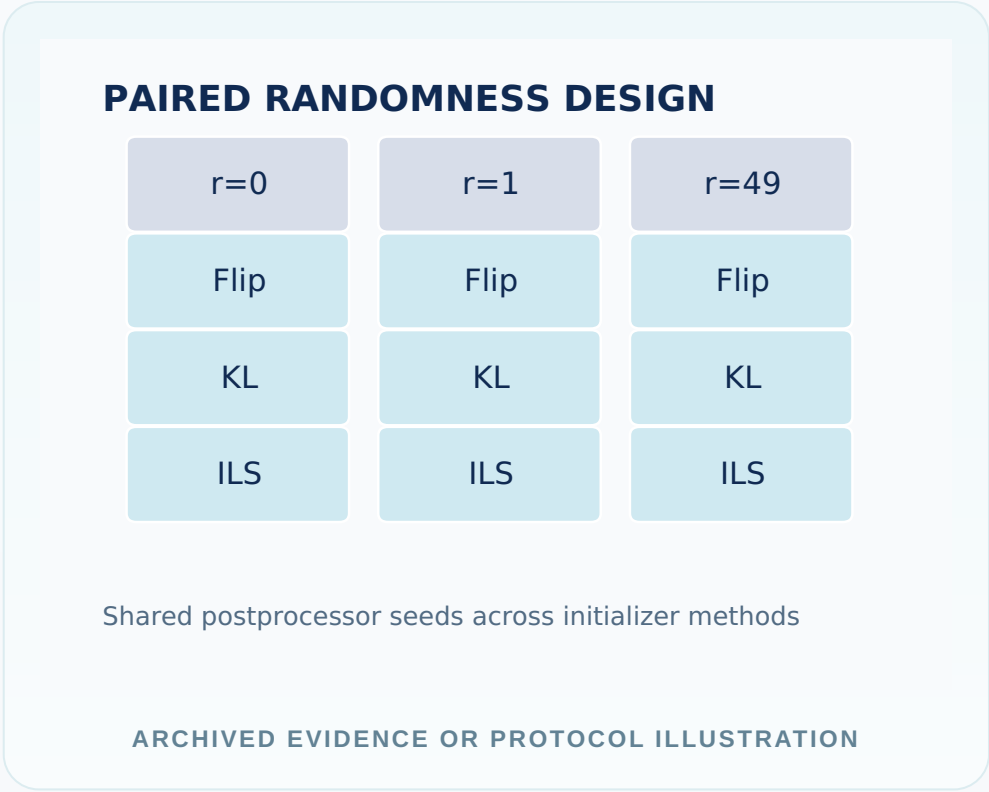
Graph index g, restart $r \in \{0, \ldots, 49\}$, seed base **2026**:

```
# Two-center rounding initialization
SeedSequence([2026, g, 2, r, 0])  # paired control
SeedSequence([2026, g, 2, r, 1])  # alternative methods

# Same postprocessing streams across methods, for matched r
flips_seed = [2026, g, r, 1313]
kl_seed    = [2026, g, r, 101]
ils_seed   = [2026, g, r, 1401]
```

Matched restart IDs control the **postprocessor randomness**, not the actual random initial centers drawn by different methods.

Source: Experiment019 and Experiment021 notes.

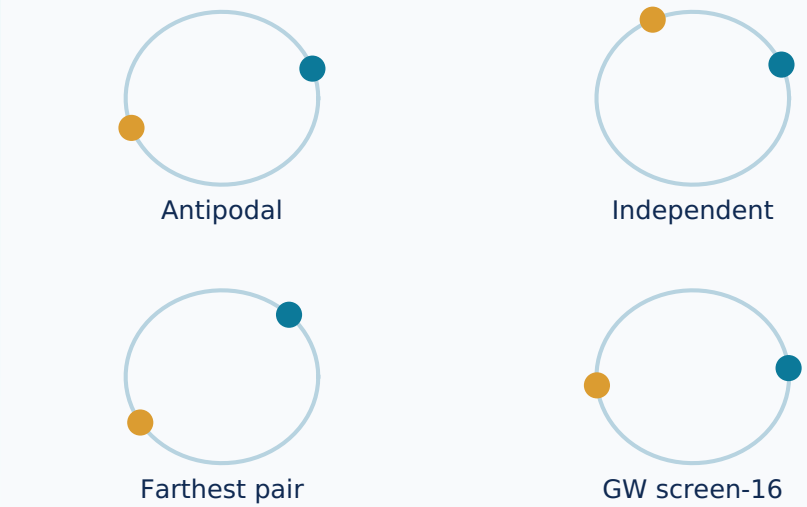


Experiment019 — Controlled pilot on G63

- Original G63 graph: **7,000 vertices**; fixed EVA factor with **119 columns**.
- **9 × 50 = 450** matched, full four-stage searches.
- Best new Experiment019 cut: **26,906** (farthest pair).
- Historical verified feasible record: **26,917** (Experiment013).
- Exploratory target: **27,047**, not a certified optimum.

Interpretation: the initializer pilot did not improve the historical G63 record, but gave a well-controlled design suitable for all G-set graphs.

Source: Experiment019 note; `RESEARCH_HANDOFF.md`.



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4. From one graph to all 71

Windows self-hosted GitHub Actions, resumability and exact coverage checks.

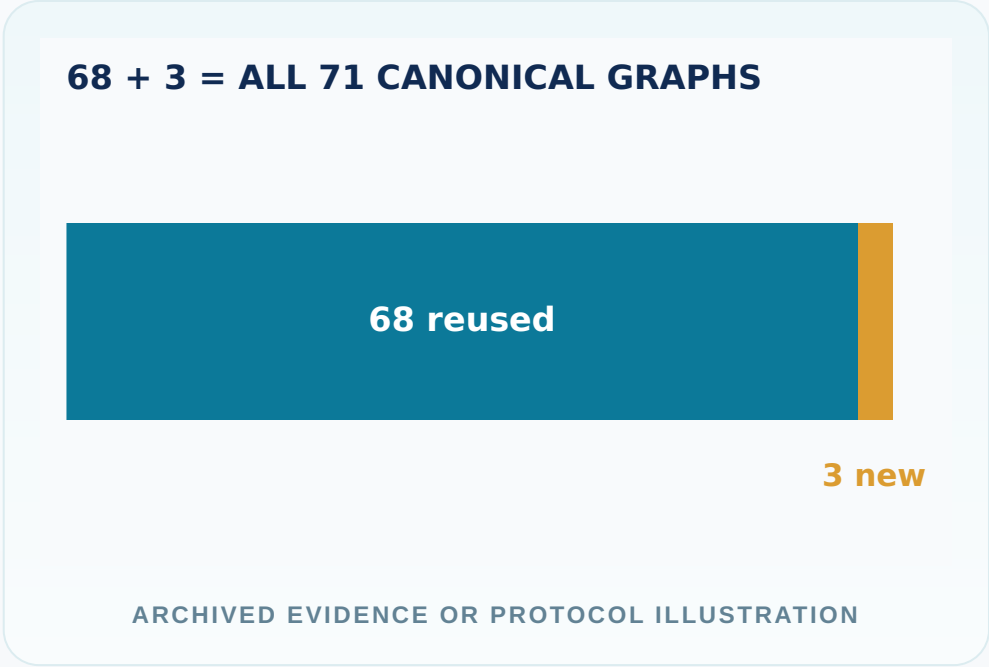
Experiment020 — Expand to the 68 graphs with saved factors

Goal: reuse already completed EVA embeddings; do not rerun the expensive factor solves.

- **68** eligible graphs with verified factors: **67 new** + reused G63.
- **68 × 9 × 50 = 30,600** archived full searches.
- **G60, G61, G62** initially excluded because matching completed factors were missing.
- **32** parallel Windows worker processes for rounding/refinement.

A failed first workflow exposed a Windows worker race while writing JSON summaries. Recovery moved per-graph publication **after worker completion** and reused intact saved graph chunks.

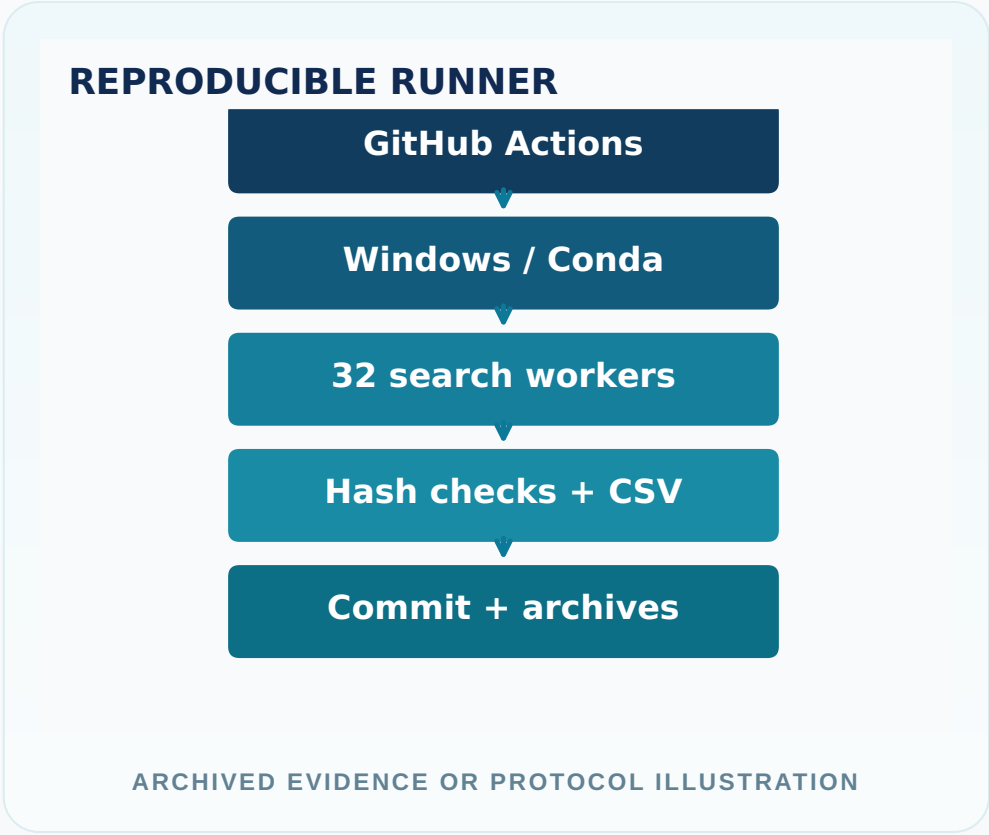
Sources: Experiment020 report, recovery run.



Why the self-hosted runner mattered

```
GitHub Actions dispatch / dedicated research branch
                  ↓
Windows 11 runner + maxkcut Conda Python
                  ↓
Factor/hash checks + fast regression tests
                  ↓
32 independent search workers; resumable chunks
                  ↓
Post-join aggregation and exact archive verification
                  ↓
Commit + push to experiment branch; upload artifact
```

Runner usage did not itself guarantee correct data: input hashes, process synchronization, restart completeness, seed identity and independent rescore checks were all required.



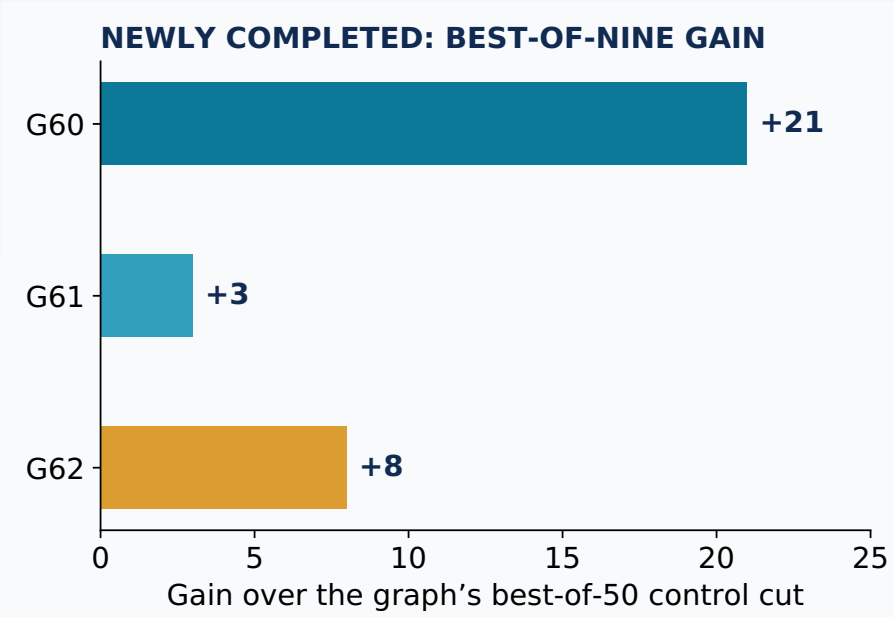
Experiment021 — Generate exactly the missing EVA factors

Graph	Vertices	Edges	EVA sweeps	Recorded state
G60	7,000	17,148	239	converged
G61	7,000	17,148	235	converged
G62	7,000	14,000	571	converged

Only these three new nonconvex factors were generated. Original Experiment019/020 factor files and search results remained unchanged.

Then: 3 × 9 × 50 = **1,350** new Round→Flip→KL→ILS searches.

Source: Experiment021 note.



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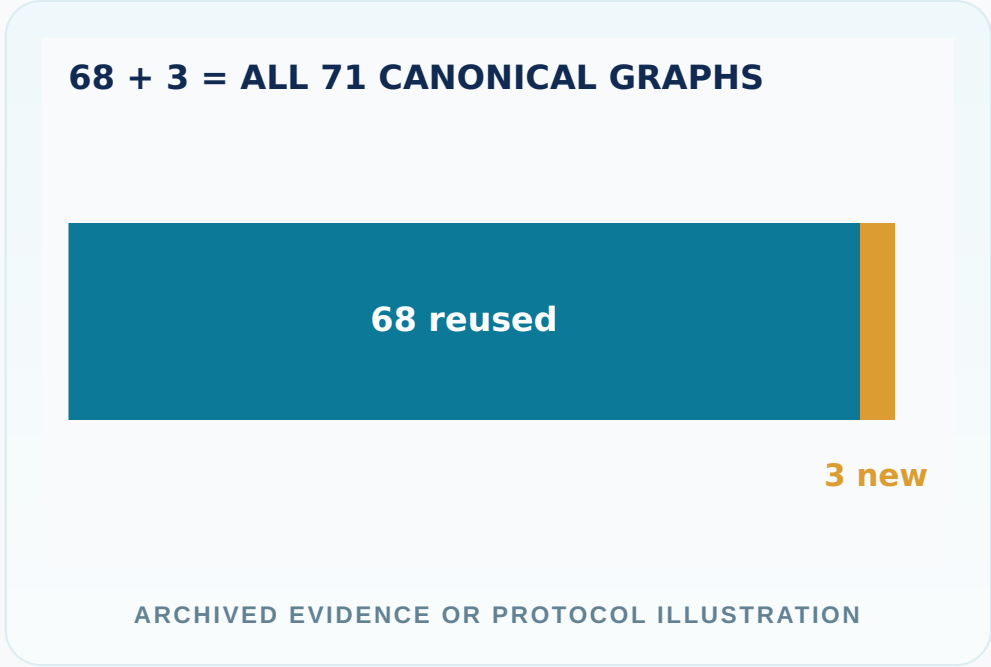
Final completeness proof is a machine-checked ledger

$$\underbrace{68 \times 9 \times 50}_{30{,}600\ \text{reused}} + \underbrace{3 \times 9 \times 50}_{1{,}350\ \text{new}} = \boxed{31{,}950}.$$

The consolidated report requires:

- Exactly **71** canonical graph identities.
- Exactly **nine** methods and **50** restart IDs per graph.
- **Zero** missing factors and **zero** incomplete instances.
- Original graph and factor **SHA-256** identities intact.
- Complete saved scores, partition files and per-restart traces.

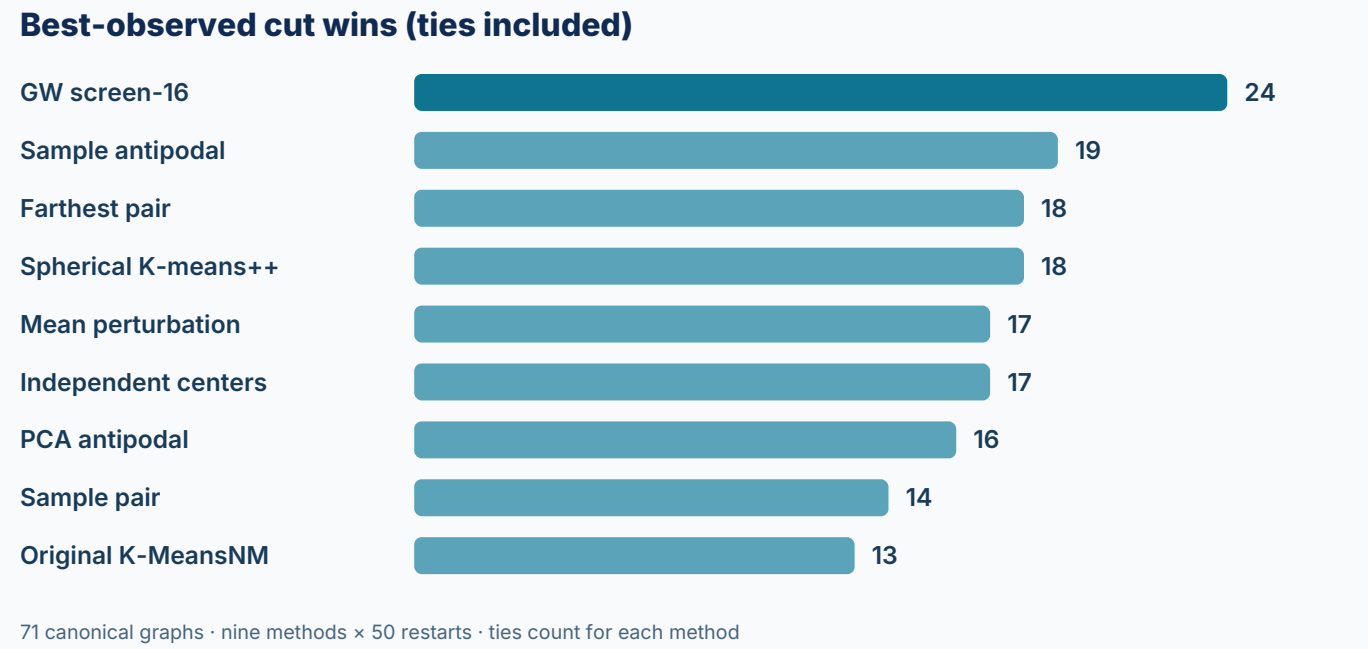
Passed: GitHub Actions #40831249367.



5. Results and interpretation

Distinguishing best individual cuts from average performance.

Ranking A — Who reached the best cut on most graphs?



Best-cut ties count for all tied methods. Values are graph counts out of 71; they need not sum to 71.

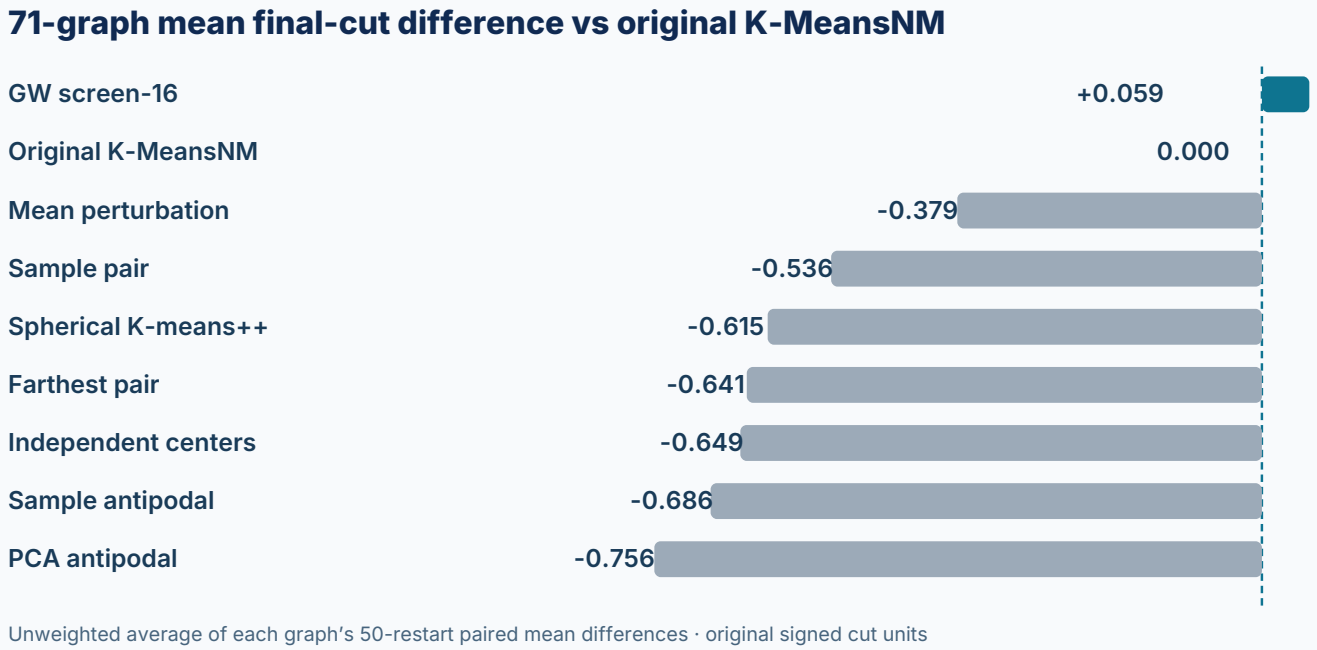
Source: `statistics/per_instance_methods.csv`, recomputed from `best_final`.

Ranking B — Share credit when several methods tie

Rank	Initialization	Graphs best (incl. ties)	Sole best	Fractional wins
1	**GW screen-16**	**24**	**7**	**12.19**
2	Farthest pair	18	**7**	9.46
3	Sample antipodal	19	5	8.75
4	Mean perturbation	17	6	8.50
5	Independent centers	17	4	7.21
6	Sample pair	14	4	6.82
7	Spherical K-means++	18	2	6.25
8	Original K-MeansNM	13	4	5.92
9	PCA antipodal	16	3	5.91

Fractional win = $1/t_g$ for each best-scoring method on graph g, where t_g is the number tied for best. The fractional wins sum to **71**.

Ranking C — Average final-cut difference versus control



Method: calculate each graph's 50-restart mean difference relative to original K-MeansNM; then average the **71 graph-level differences** equally. Positive is better.

Most alternatives are slightly below the control. GW screen-16 is **+0.059 raw cut units**, with extra evaluations and no established statistical dominance.

Three newly completed instances: best observed cuts

Graph	Best initializer	Best of nine	Control best	Gain
G60	Sample antipodal	**14,124**	14,103	**+21**
G61	Independent centers	**5,721**	5,718	**+3**
G62	Mean perturbation	**4,816**	4,808	**+8**

All values are **feasible signed Max-Cut scores**, not certificates of optimality.

Source: Experiment021 finalized per-graph records.

Important graph-specific landmarks

G63

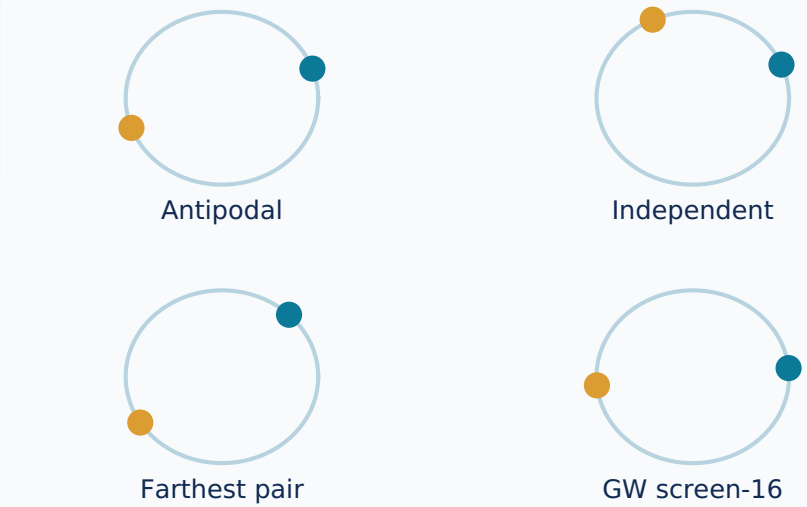
- Historical best observed feasible cut: **26,917** (Experiment013).
- Experiment019's nine-initializer best: **26,906**.
- Distance from historical record to exploratory **27,047** target: **130**.

G11

- A search attained **564** on the canonical instance, matching the repo's cited established optimum for that weighted graph.

Do not transfer historical targets, cut scales or certified-optimum labels between different graph identities.

Sources: `RESEARCH_HANDOFF.md`, Experiment019 note, `docs/gset-reference-values.md`.



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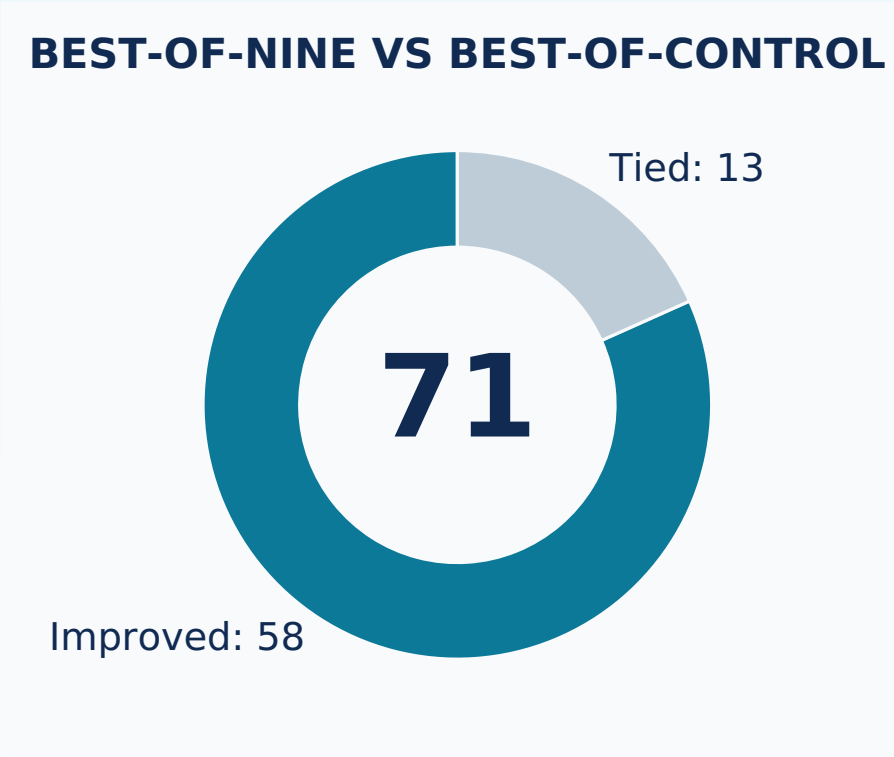
Why 58/71 is not an equal-budget victory

The best of nine methods beat the best of 50 control restarts on **58 graphs** and tied on **13**.

But it had up to **450 searches per graph**, versus **50** for the original K-MeansNM alone.

A portfolio containing the control can never score worse in best-of-search, by construction. Its 58 victories are empirical evidence of *diversity and opportunity*, not proof that an individual method is more effective per unit computation.

Also, GW screen-16 considers 16 additional initial hyperplanes per restart.



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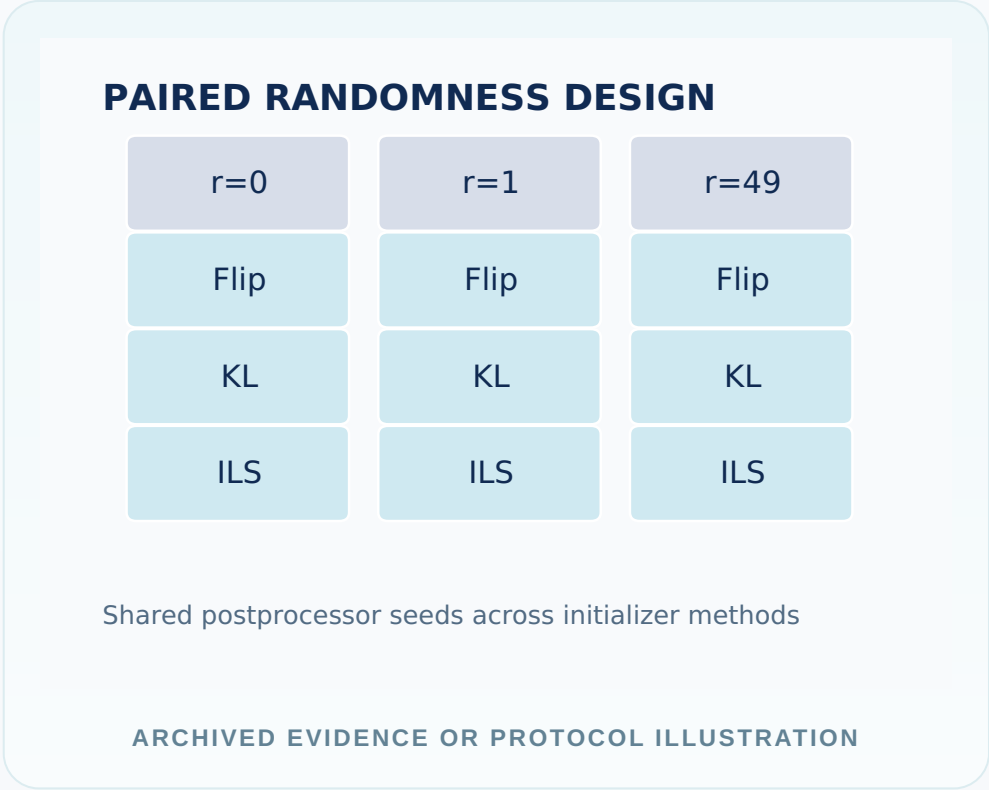
What the statistics do—and do not—show

Paired comparison at graph g:

$$\Delta_{g,m} = \frac{1}{50} \sum_{r=0}^{49} \left[C^{\text{final}}_{g,m,r} - C^{\text{final}}_{g,\text{control,r}} \right].$$

- Report the mean of the **71** graph-level $\Delta_{g,m}$ values.
- Within graph, **4,000-replicate paired bootstrap** intervals examine restart variability.
- Do **not** treat 31,950 correlated restarts as independent graphs.
- Do **not** equate a best-of-50 score with expected performance.
- No universal approximation ratio follows from these cut scores.

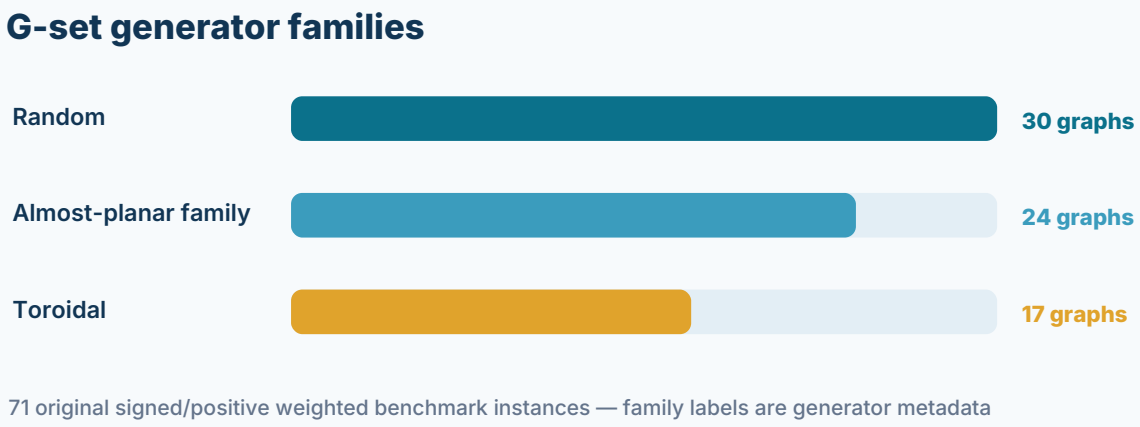
Source: Experiment021 note; `paired_vs_original_kmeansnm.csv`.



6. Does graph topology explain the best initializer?

Experiment022 · retrospective analysis of the completed 71-graph data

The G-set is not one homogeneous population



G-set generator families
Random
30 graphs
Almost-planar family
24 graphs
Toroidal
17 graphs
71 original signed/positive weighted benchmark instances — family labels are generator metadata

ARCHIVED EVIDENCE OR PROTOCOL ILLUSTRATION

Benchmark generator families (secondary metadata):

- **Random:** 30 graphs.
- **Planar-family / almost-planar:** 24 graphs.
- **Toroidal:** 17 graphs.

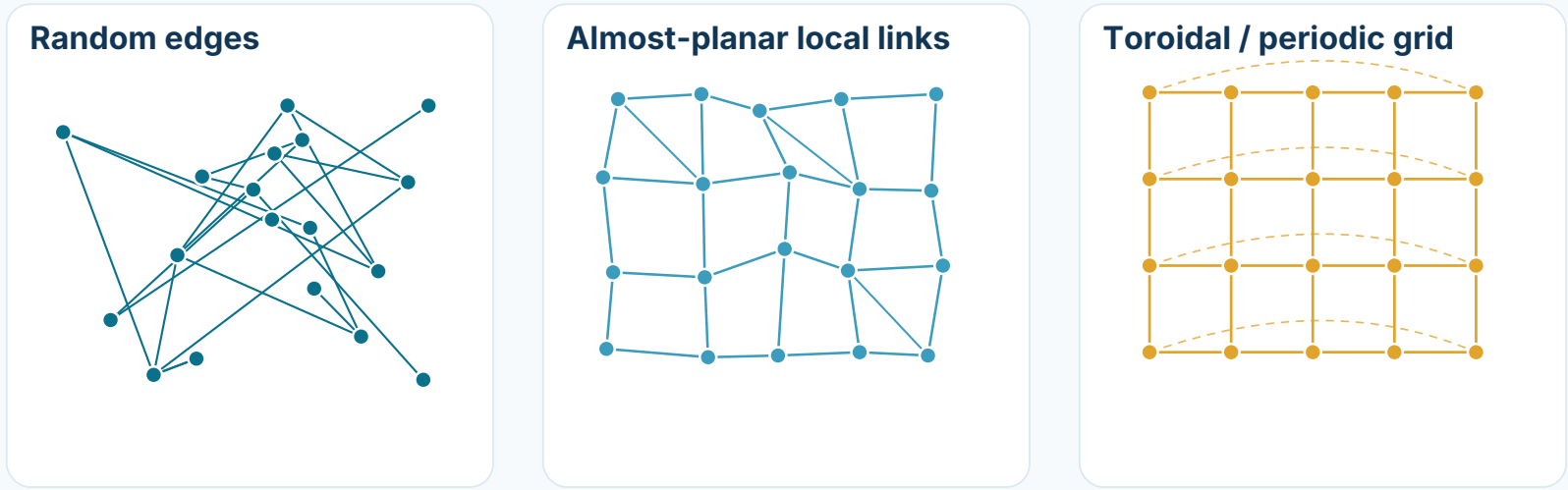
These categories are useful explanatory variables, but they are **not independent random samples of all possible graphs**.

Original inputs supply **edges and weights**, not canonical vertex coordinates.

Sources: `notes/experiments/experiment022-graph-structure-methods.md`; OpenIsing G-set metadata.

Graph drawing is not graph topology

Connectivity patterns — schematics, not benchmark coordinates



Topological features must come from original edges, not the drawing algorithm

The illustrations are **schematic**. A spring-layout ring, PCA cloud or low-dimensional EVA projection does **not** prove a graph is spatially circular.

Measure connectivity from the **original adjacency lists**, and independently measure **EVA vector geometry** from the frozen factors.

Real structural signatures: four audited examples

Structural signatures measured from original edges

Graph	Generator family	Mean degree	Degree CV	Transitivity
G1	random	47.94	0.132	0.059
G14	planar-family	11.74	0.983	0.154
G11	toroidal	4.00	0	0
G48	toroidal	4.00	0	0

G14 exceeds the strict planar graph edge bound — "planar-family" is a benchmark label

- **Degree coefficient of variation** captures irregularity.
- **Transitivity** measures the fraction of connected triples closed into triangles.
- **G11 and G48** are degree-four and triangle-free; short **four-cycles** and signed frustration may be more diagnostic.

The published **planar-family** label is not a proof of mathematical planarity. G14 has more than $3n - 6$ edges.

Experiment022 — Winners change across graph families

Best observed cut: number of graphs won or tied

	Random · 30	Planar family · 24	Toroidal · 17
Original paired	7	2	4
Independent	8	5	4
Sample-antipodal	10	4	5
Sample-pair	5	5	4
Farthest-pair	9	3	6
Spherical++	7	4	7
Mean-perturb	7	4	6
PCA-antipodal	7	3	6
GW screen-16	5	8	11

All ties count. GW screen-16 performs 16 extra trial-cut evaluations per restart.

Best of 50 starts, after the same refinement protocol:

- Random: **sample-antipodal**, best/tied on **10/30**.
- Planar-family: **GW screen-16**, **8/24**.
- Toroidal: **GW screen-16**, **11/17**.

Several methods tie. The table counts **all tied winners**—it is not a set of mutually exclusive victories.

Source: Experiment022 `method_by_family.csv`.

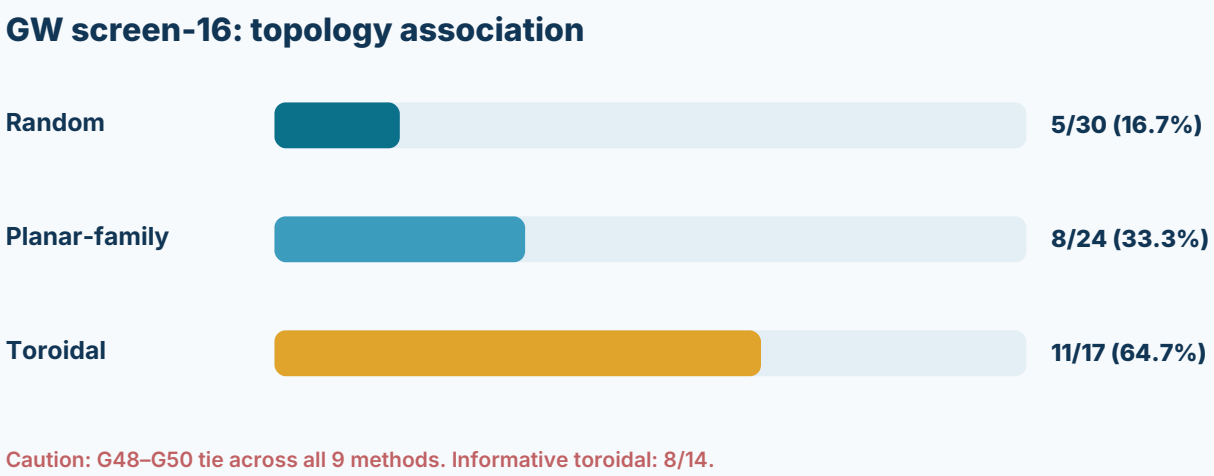
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All ties count. GW screen-16 performs 16 extra trial-cut evaluations per restart.

ARCHIVED EVIDENCE OR PROTOCOL ILLUSTRATION

Screened hyperplanes: strong descriptive topology association



GW screen-16 ties or wins the highest cut on **5/30 random**, **8/24 planar-family**, and **11/17 toroidal** graphs.

But G48–G50 are **uninformative**: all nine methods reach the same best cut.

Excluding those three: **8 of 14 informative toroidal graphs**.

Budget warning: screening uses **16 additional initial cut evaluations per restart**. The difference is not a fair, equal-time method comparison.

GW screen-16: topology association
Random
5/30 (16.7%)
Planar-family
8/24 (33.3%)
Toroidal
11/17 (64.7%)
Caution: G48–G50 tie across all 9 methods. Informative toroidal: 8/14.

ARCHIVED EVIDENCE OR PROTOCOL ILLUSTRATION

A high win frequency does not imply a high mean cut

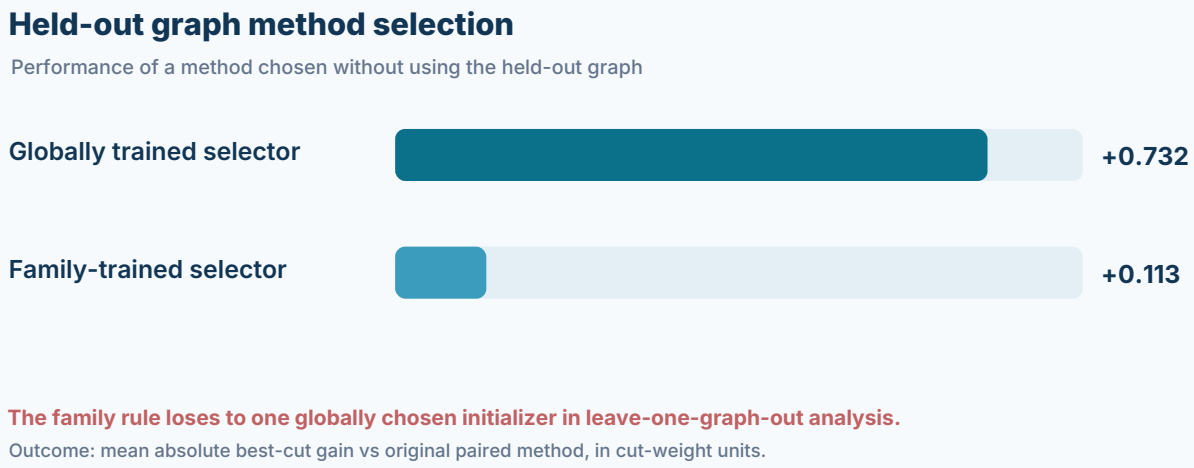
Graphwise mean final-cut difference versus paired K-MeansNM:

Initializer	Random (30)	Planar-family (24)	Toroidal (17)
GW screen-16	+0.115	**−0.399**	**+0.605**
Sample-antipodal	−0.374	−1.196	−0.518
Mean perturbation	−0.615	−0.671	**+0.452**
Original paired	0	0	0

These are **raw cut-weight units**, not percentages. Best-of-50 wins and 50-run means answer **different questions**.

Source: Experiment022 `method_by_family.csv`.

Can we select a method for an unseen graph?



Held-out graph method selection
Performance of a method chosen without using the held-out graph
Globally trained selector
+0.732
Family-trained selector
+0.113
The family rule loses to one globally chosen initializer in leave-one-graph-out analysis.
Outcome: mean absolute best-cut gain vs original paired method, in cut-weight units.

ARCHIVED EVIDENCE OR PROTOCOL ILLUSTRATION

We evaluated a **leave-one-graph-out** rule:

1. Exclude the held-out graph entirely.
2. Choose a method from the other 70 graphs.
3. Test its best-of-50 cut on the held-out graph.

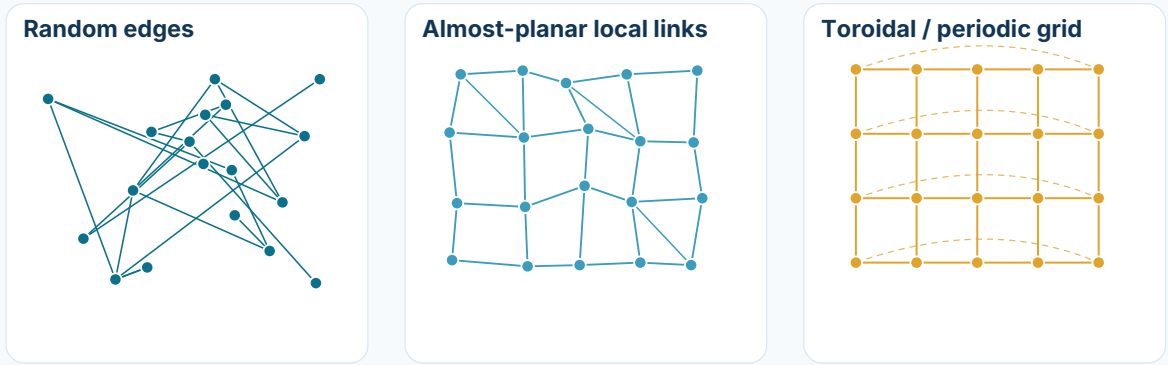
Mean held-out gain vs paired:

- Global method selection: **+0.732**.
- Family-specific selection: **+0.113**.

Conclusion: the simple family rule did **not** outperform one globally chosen method. The descriptive association is not yet a predictive algorithm.

Why topology alone is insufficient

Connectivity patterns — schematics, not benchmark coordinates



Topological features must come from original edges, not the drawing algorithm

Connectivity patterns — schematics, not benchmark coordinates



Topological features must come from original edges, not the drawing algorithm

ILLUSTRATIVE GRAPH GEOMETRY

Possible confounders are intertwined with graph family:

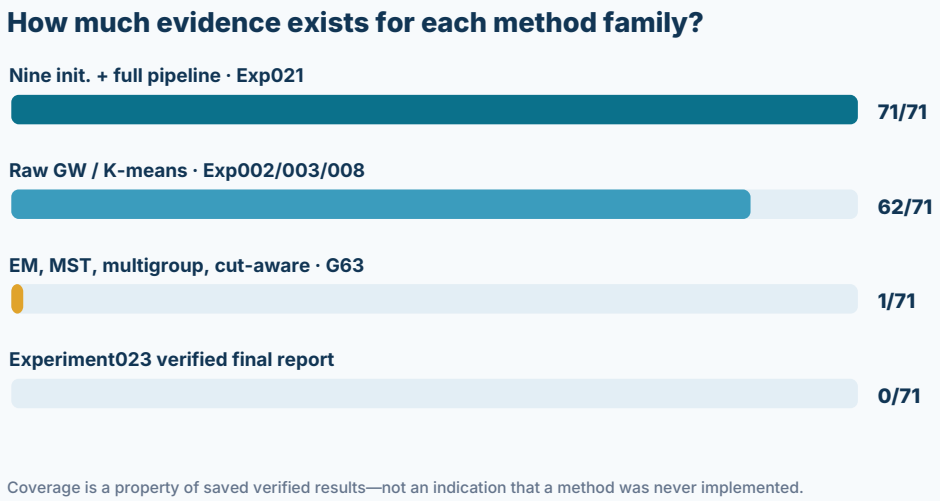
- Number of vertices, edges and degree distribution.
- Positive-only versus **signed ±1** weights.
- Cycles, frustration, components, connectivity and spectral gap.
- Nonconvex EVA factor geometry and effective rank.
- Generator siblings and repeated graph designs.

Next test: a pre-registered selector using structural features, evaluated with **held-out graph families**, and with equalized compute.

7. What about the older MST, EM and coarsening methods?

The historical study gives **additional evidence**, but its coverage and search budgets differ.

Audit the amount of actual comparable evidence



How much evidence exists for each method family?
Nine init. + full pipeline · Exp021
71/71
Raw GW / K-means · Exp002/003/008
62/71
EM, MST, multigroup, cut-aware · G63
1/71
Experiment023 verified final report
0/71
Coverage is a property of saved verified results—not an indication that a method was never implemented.

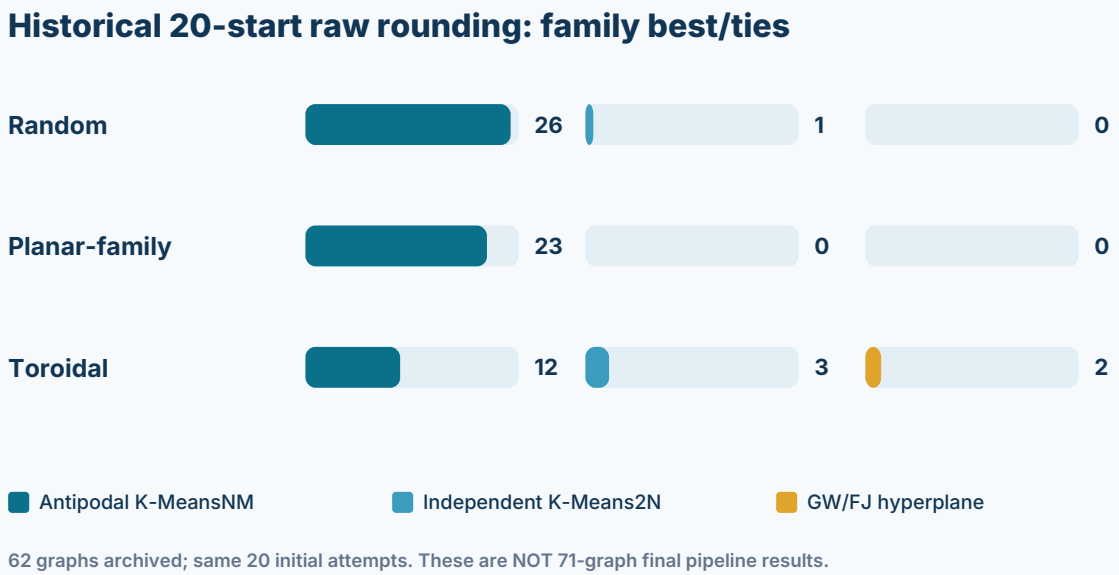
ARCHIVED EVIDENCE OR PROTOCOL ILLUSTRATION

Three evidence strata (do not pool):

1. **71 graphs:** nine Experiment021 initialization methods, complete Round → Flip → KL → ILS.
2. **62 graphs:** archived **20-start raw rounding** by GW/FJ, K-Means2N and antipodal K-MeansNM.
3. **G63 only:** MST, soft EM/CEM/Gibbs, multi-group K-means, cut-aware and specialized KL/ILS trials.

Experiment023: 17 registered protocols; **no verified complete numerical report** found on its branch at deck preparation.

Historical raw rounding: the antipodal control is strong



From the archived **62-graph**, **20-start** raw rounding comparison:

- Random: antipodal K-MeansNM wins/ties on **26/27**.
- Planar-family: **23/23**.
- Toroidal: **12/12**.

Overall, it wins or ties on **61/62** graphs among these three algorithms.

How much did antipodal K-means improve over raw GW?

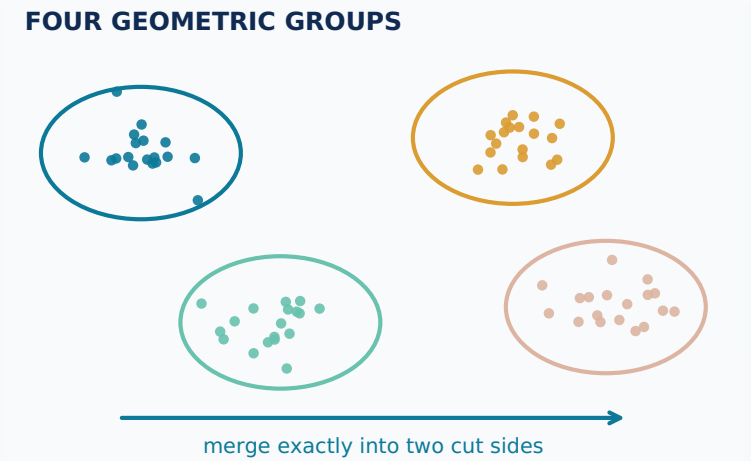
At the **best-of-20 rounding stage**, unweighted mean within-family relative improvements were:

Family	Graphs with archived raw results	Mean relative K-MeansNM gain vs GW/FJ
Random	27	**+2.032%**
Planar-family	23	**+1.723%**
Toroidal	12	**+1.019%**

These are descriptive, graph-normalized **best-of-20** comparisons, not approximation guarantees or global significance tests.

Different restart and refinement budgets prevent direct numerical pooling with Experiment021.

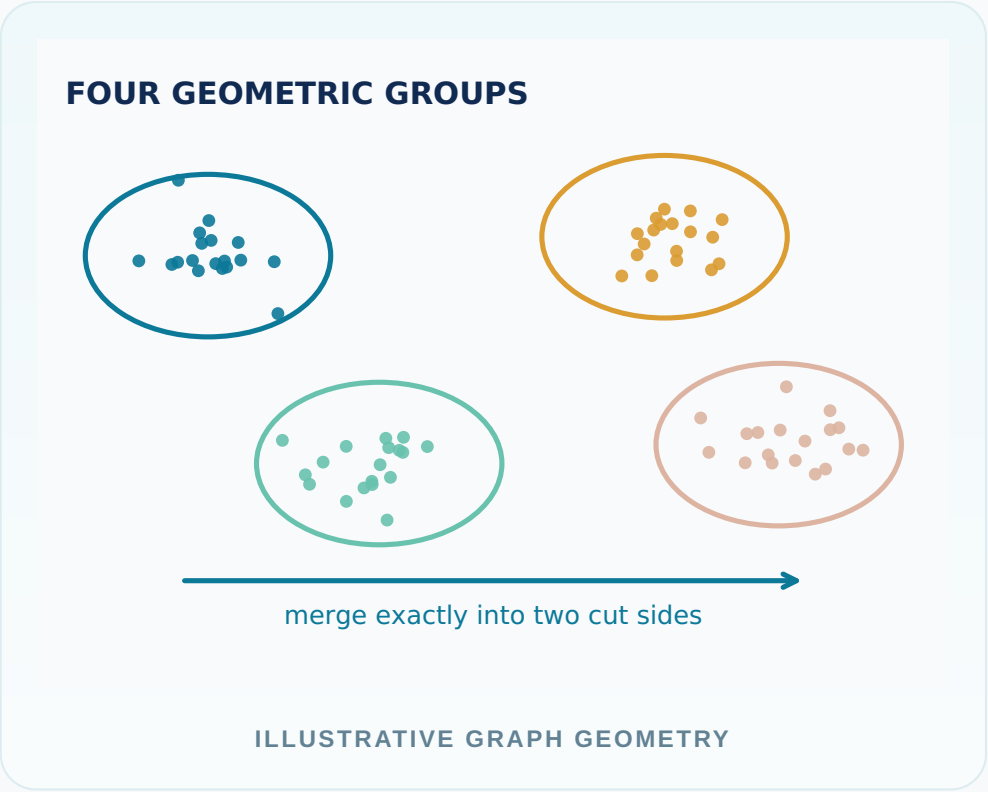
Exact MSTs in the high-dimensional EVA space



1. Compute an exact MST over **all EVA vectors** in $\mathbb{R}^p$; metric: L1, L2 or L∞.
2. Remove one edge (**longest**, **balanced**, **diameter center** or **best original cut**) or several edges.
3. Merge components into two **binary** cut sides.
4. Evaluate signed cut and optionally refine original vertices.

Important: the MST is built from **vector distances**, not the signed graph weights.

Cost: roughly $O(n^2 p)$ for each exact full-space MST; large graphs need explicit feasibility checks.



Why one MST edge is not always enough

For a tree edge e, the resulting cut can be written as

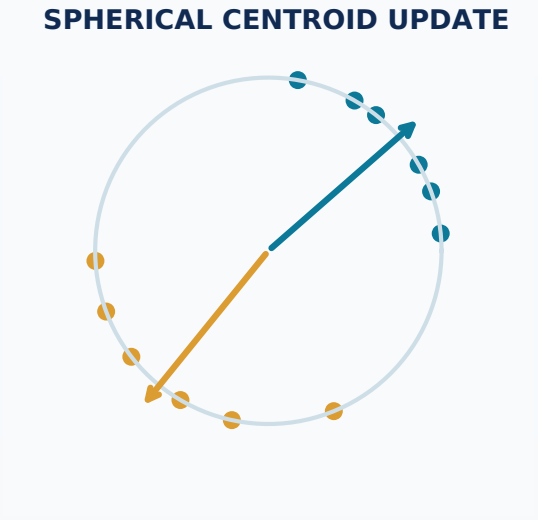
$$C(e) = \sum_{\{i,j\} \in E} w_{ij}\, \mathbf{1}[e \in P_T(i,j)],$$

where $P_T(i,j)$ is the unique path between i and j on the **geometric MST**.

- Longest edge optimizes **distance**, not signed cut.
- **Best-cut edge** explicitly scores all $n-1$ possible one-edge tree splits.
- Multi-edge coarsening optimizes over a contracted **signed** graph.

This suggests a structural hypothesis: **MST performance should depend on how EVA-space geometry aligns with graph-edge frustration**.

Soft EM-like rounding versus hard K-means

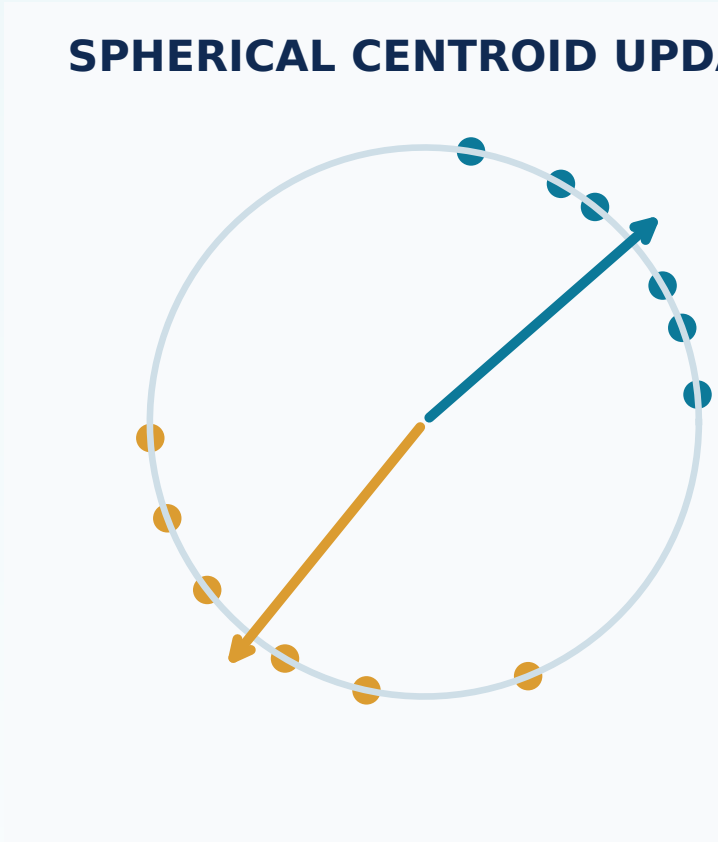


Instead of hard cluster membership, assign $p_i \in [0, 1]$ and optimize a **mean-field objective** combining:

- Expected **signed Max-Cut** under independent Bernoulli labels.
- Geometric agreement with EVA centroids.
- Entropy regularization / temperature annealing.

Then generate hard candidates through **thresholding, CEM or Gibbs**, with optional KL refinement.

Not standard maximum-likelihood EM: the repository calls this an **EM-like variational alternating** procedure.



ILLUSTRATIVE GRAPH GEOMETRY

The elite-source trap: retaining is not improving

G63 historical record: 26,917 was already achieved by the Experiment013 K-MeansNM + Flips + KL + ILS pipeline.

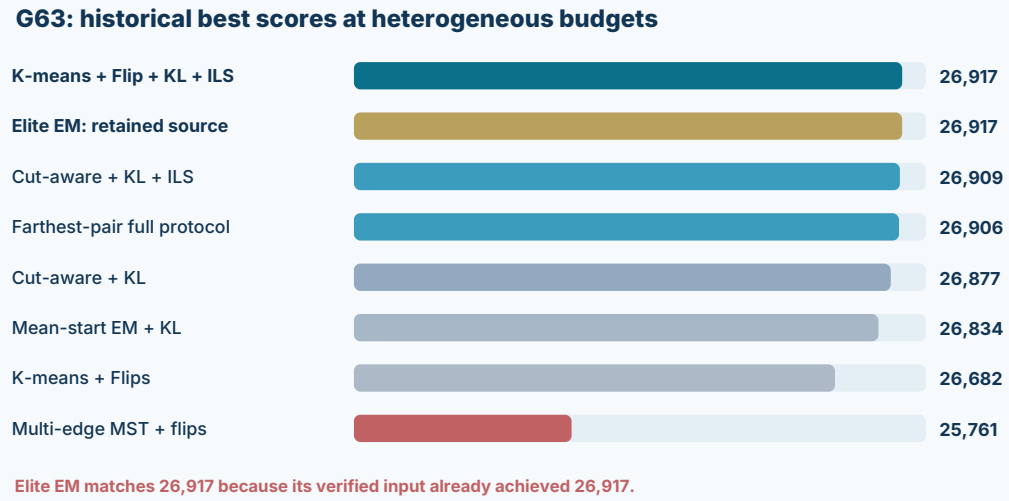
Experiment016's **elite-initialized EM/CEM/Gibbs** variants also reported **26,917**.

However, their starting **source partition already achieved 26,917**, and none of those elite episodes improved its source cut.

Interpretation: matching the historical record by retaining an elite is **not** evidence that EM independently discovered that solution.

The new Experiment023 protocol deliberately excludes elite warm starts from its primary EM arms.

G63 case study: cross-method scores, different budgets



G63: historical best scores at heterogeneous budgets
K-means + Flip + KL + ILS 26,917
Elite EM: retained source 26,917
Cut-aware + KL + ILS 26,909
Farthest-pair full protocol 26,906
Cut-aware + KL 26,877
Mean-start EM + KL 26,834
K-means + Flips 26,682
Multi-edge MST + flips 25,761
Elite EM matches 26,917 because its verified input already achieved 26,917.

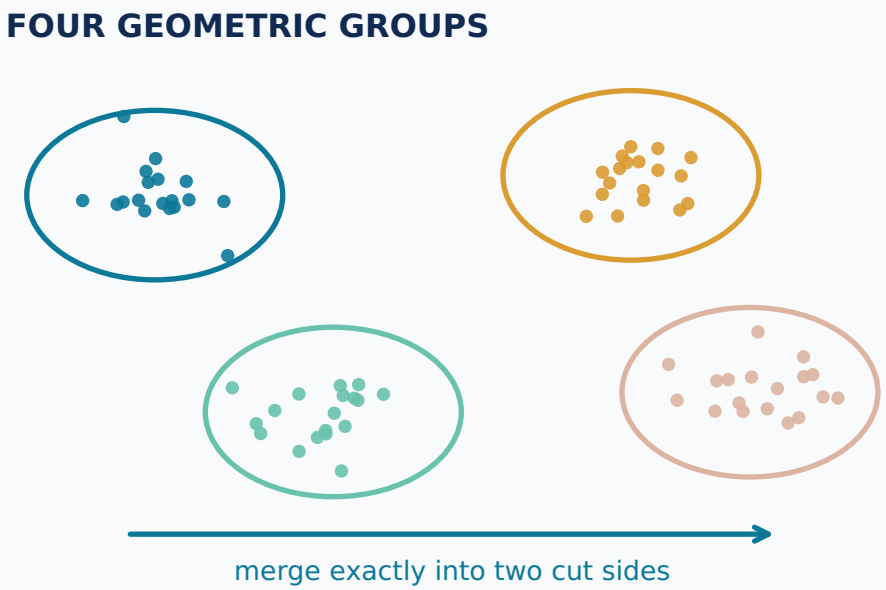
ARCHIVED EVIDENCE OR PROTOCOL ILLUSTRATION

The strongest observed cut remains **26,917** (Experiment013).

- Cut-aware + KL + ILS: **26,909**.
- Farthest-pair full pipeline: **26,906**.
- Mean-start EM + KL: **26,834**.
- Multi-edge MST + vertex flips: **25,761**.

Do not rank them as if they had equal attempts or identical postprocessing. The figure is a verified historical **archive comparison**, not a controlled tournament.

Multigroup K-means: clustering k ≠ Max-k-Cut



For temporary k-way EVA clusters, aggregate original signed graph edges into W_{ab}.

Then find the best **binary merge**:

$$C(z) = \sum_{a<b} W_{ab}\mathbf{1}[z_a \neq z_b], \quad z_a \in \{0,1\}.$$

Historical data cannot settle a topology-wide MST/EM ranking

Method family	Archived structural coverage	Biggest interpretive issue
GW/FJ, K-Means2N, K-MeansNM raw	**62** graphs	Different stage than Exp021
Nine initializers + common refinement	**71** graphs	GW screen-16 screening overhead
MST, EM, cut-aware, multi-group and advanced search	**G63** chiefly	Different pilots, budgets and stages
Direct hyperplane / selected older studies	Historical code/notes	Some consolidated summaries unverified

The missing scientific experiment is not another chart. It is a **new, prospective, consistent benchmark** across graph families.

8. Experiment023 — a prospective all-method benchmark

Registered design · implementation committed · completion not independently verified

From initializer tournament to a family-wide benchmark

Experiment023 — 17 registered protocols

Control	Hyperplane / cut-aware	Multi-group	Soft EM / sampling	MST v
2 protocols	3 protocols	2 protocols	3 protocols	7 prot

All methods → SAME 20 Flips → 30 KL passes → 80 ILS cycles

Exception: Tabu control changes the refiner. MST preprocessing is costly/deterministic.

This slide shows the registered design; it is not a graph of completed outcomes.

Experiment021 asked: *Which centroid initialization works best?*

Experiment023 asks: **Which whole rounding family yields the strongest cut under a shared refinement policy and bounded computation?**

17 registered protocols × **71 original graphs** = **1,207 planned graph-method cells**.

Only **verified completed cells** can later enter an empirical ranking.

Experiment023 — 17 registered protocols

Control 2 protocols | Hyperplane / cut-aware 3 protocols | Multi-group 2 protocols | Soft EM / sampling 3 protocols | MST v 7 prot

All methods → SAME 20 Flips → 30 KL passes → 80 ILS cycles

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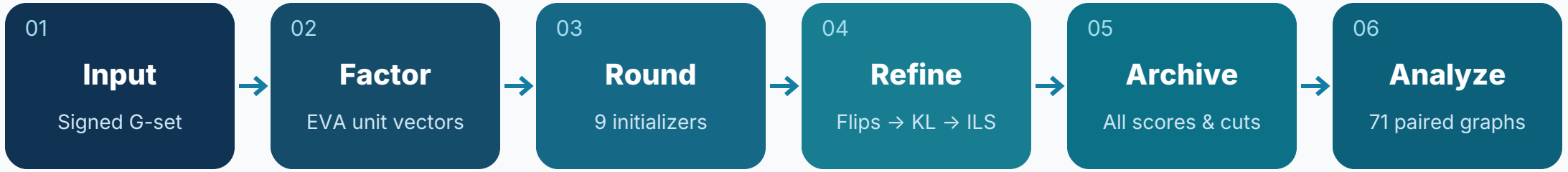
ARCHIVED EVIDENCE OR PROTOCOL ILLUSTRATION

The 17 Experiment023 registered protocols

Rounding / optimization family	Protocols
K-means controls	`control-paired`, `control-tabu`*
Hyperplane / cut-aware	`gw-threshold`, `cutaware-kmeans`, `hyperplane-sweep`
Multi-group EVA clustering	`multigroup-k4`, `multigroup-k8`
EM-like and stochastic	`em-hard`, `em-cem`, `em-gibbs`
Exact L2 MST	`mst-l2-longest`, `-balanced`, `-diameter`, `-bestcut`, `-multicut`
Exact alternative-distance MST	`mst-l1-bestcut`, `mst-linf-bestcut`

*Tabu changes the **refiner**, not merely an initializer; analyze separately.*

The full-run common downstream pipeline



One fixed vector factor per graph · 50 matched restarts · immutable saved partitions

For **comparable hard binary seeds**, use:

Round → up to 20 positive flip sweeps → 30 generalized KL passes → 80 ILS cycles (eight inner KL passes; kicks 8/16/32/64).

Maximum 100 restarts or a 240-second per-method wall-time budget, whichever comes first, with hard timeout at 510 seconds.

MST is deterministic per factor and metric; **do not pretend repeated identical trees are independent restarts**.

The fairness protocol: control what is comparable

One graph, one frozen source factor, same original signed cut objective.

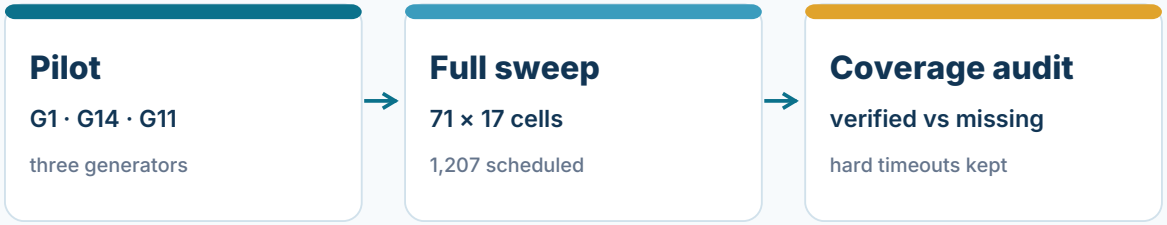
Record separately:

1. **Raw** candidate score before refinement.
2. Flip, KL and ILS **final** scores.
3. Factor/graph SHA, preprocessing cost, evaluations and wall time.
4. Warm-start or elite source (if any), seeds and trace.
5. **Timeout**, failure, budget-limited or completed status.

The 240-second target is a **ceiling**, not a mathematically exact equal-work guarantee. MST preprocessing and EM sampling may dominate the cost.

Pilot gate → complete sweep → coverage audit

Experiment023: pilot-gated run and verification



Stop condition: if pilot fails, do not claim the 71-graph run commenced.

Persist per-case graph/factor SHA, stage cuts, runtime, seed and failure status.

Experiment023: pilot-gated run and verification

Pilot
G1 · G14 · G11
three generators
Full sweep
71 × 17 cells
1,207 scheduled
Coverage audit
verified vs missing
hard timeouts kept

Stop condition: if pilot fails, do not claim the 71-graph run commenced.

Persist per-case graph/factor SHA, stage cuts, runtime, seed and failure status.

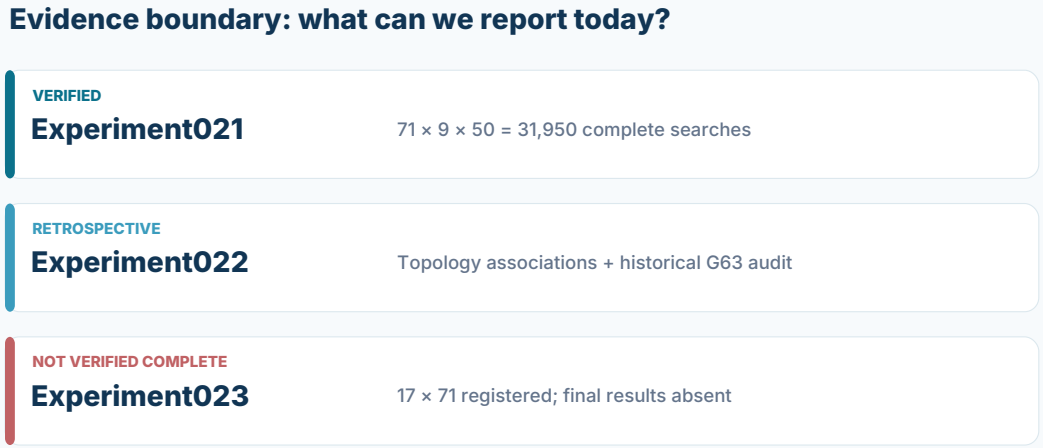
ARCHIVED EVIDENCE OR PROTOCOL ILLUSTRATION

Pilot graphs: G1 (random), G14 (planar-family), G11 (toroidal).

- Validate code, factor hashes, feasible cuts and computational costs.
- Full workflow starts **only if every pilot succeeds**.
- One self-hosted Windows runner; checkpoint one graph at a time.
- Audit a **71 × 17 = 1,207-cell** outcome matrix.

A declared cell is not a completed run. A timeout must remain visible in the final report.

Evidence status: separate proven data from the planned study



At preparation, the Experiment023 branch contained its workflow, methods and protocol but **no verified final coverage/report JSON**.
Therefore, this presentation gives no invented Experiment023 winner, score, speedup or topology-specific conclusion.
Link: Experiment023 protocol.

How to analyze Experiment023 when results exist

Primary question: within each original graph and a bounded budget, which method returns the best *verified feasible* cut?

A prespecified analysis should include:

- **Within-graph normalized gain** against new paired control.
- Fractional wins with ties split across winning protocols.
- Runtime-quality **Pareto frontier**; shared MST cache cost allocation.
- Separate positive-only and signed-weight graphs.
- Random / planar-family / toroidal **stratification**.
- Leave-generator-family-out validation and graph-level intervals.

Do not pool 1,207 correlated cells as 1,207 independent graph samples.

9. What we learned — and what remains open

A scientific story of **reproducible experimentation**, not a single champion score.

The full research trajectory, one view



Early program: factor construction, GW/K-means, exact MST, KL/Tabu, multigroup and soft EM.
Experiment021: 71 verified graphs, 31,950 initializations/refinements.
Experiment022: topology and historical-coverage retrospective audit.
Experiment023: prospective all-family protocol designed; final run evidence still pending.

Four different questions must stay distinct

Question	Statistic	Current strongest evidence
Best *individual* cut?	Best-of-50	GW screen-16 most best/ties in Exp021
Best *typical* restart?	Graphwise mean final cut	Original paired is competitive; GW +0.059 raw units
Best method for a **graph family**?	Family-stratified wins	Descriptive GW toroidal association; not predictive yet
Best method *per CPU-second*?	Equal-budget anytime cuts	**Not established** by Exp021/022; Experiment023 proposed

These distinctions prevent choosing a champion by **changing the evaluation target after seeing results**.

What can be concluded rigorously from this record?

Supported statements:

- The original graph and final feasible signed Max-Cut are independently verified.
- The G-set Experiment021 study is **complete**: 71 graphs and 31,950 searches.
- GW screen-16 achieved the most best-cut ties/wins, but spent more initial evaluations.
- Graph generator family correlates with some observed method wins.
- The simple topology selector did not beat global selection in held-out comparison.
- Historical MST/EM results cannot justify all-graph ranking without wider runs.

Not supported: new SDP optimality certificate, a Max-k-Cut ($k > 2$) claim, equal-budget superiority, or final Experiment023 winners.

Reproducibility: every claim links to an archive

Item	Source of truth
Canonical signed G-set inputs	`data/G*.txt`
Experiment021 complete cuts/restarts	`results/gset/experiment021-.../statistics/`
Experiment022 labels/LOOCV/history	`results/gset/experiment022-graph-structure-methods/`
Experiment023 registered protocol	`notes/experiments/experiment023-all-method-topology.md`
Experimental implementation	`src/maxkcut/`
Regenerable vector charts	`presentations/.../build_new_figures.py`

Retain source factor SHA, graph SHA, verified partition, configuration ID and method-stage timing in all future reports.

Immediate next scientific checkpoints

After the Experiment023 pilot and sweep:

1. Audit complete graph-method **coverage and timeouts**.
2. Re-score every archived cut and validate source-factor identity.
3. Compare **cut vs wall-time** under the same method budget.
4. Measure structural predictors: degree CV, cycles, sign fraction and EVA spectrum.
5. Evaluate structural method selectors on graph families withheld from training.
6. Report effect size, graph-level uncertainty, multiplicity and negative results.

The **method × topology** hypothesis remains open until the planned new outputs are verified.

Final synthesis

There is no single universal winner yet. We have a complete nine-initializer benchmark, new topology associations, an audited history of alternative algorithms, and a controlled plan to test them all.

Literature and permanent experiment sources

Research foundations

- Rodríguez Fernández, Á. E. (2021). *On the Study of Fitness Landscapes and the Max-Cut Problem* (Chs. 6–7).
- Rodríguez-Fernández, Á. E., et al. (2020). *Clustering Improves the Goemans–Williamson Approximation for the Max-Cut Problem*. *Computation* **8**, 75. DOI: 10.3390/computation8030075.
- Felzenszwalb, P., Klivans, C., & Paul, A. (2022). *Clustering with Semidefinite Programming and Fixed Point Iteration*. *JMLR* **23**, 21-0402.

Primary archived results and design

- Experiment021 report — full 71 graphs.
- Experiment022 note — topology and historical methods.
- Experiment023 note — future outcomes not verified.

Prepared for Professor Ángel E. Rodríguez Fernández · 29 October 2026.